FPT UNIVERSITY — HO CHI MINH CITY CAMPUS
DEPARTMENT OF SEMICONDUCTOR AND AUTOMOTIVE ENGINEERING

FPT UNIVERSITY

LAB REPORT

Lab 3 — Combinational MOS Logic Circuit

CMOS Full Adder Design

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Class	IC2007
Subject	ICI211

1 BACKGROUND THEORY

A full adder is a combinational circuit that adds three binary inputs — A, B and C_{in} — and produces two outputs: Sum (S) and Carry-out (C~out~). The Sum is formed by XOR-ing the three inputs; the Carry-out by an AND/OR majority function.

$$S = A \oplus B \oplus C_{in}$$

$$C_{out} = A{\cdot}B + A{\cdot}C_{in} + B{\cdot}C_{in}$$

The truth table on the right enumerates all eight input combinations and verifies the intended behaviour of the circuit.

A	B	C_{in}	S	C_{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

2 PART A — SCHEMATIC DESIGN

The full adder is built hierarchically from CMOS NAND and XOR primitives. Each cell was drawn at transistor level and wrapped in a symbol so it can be instantiated in the top-level schematic.

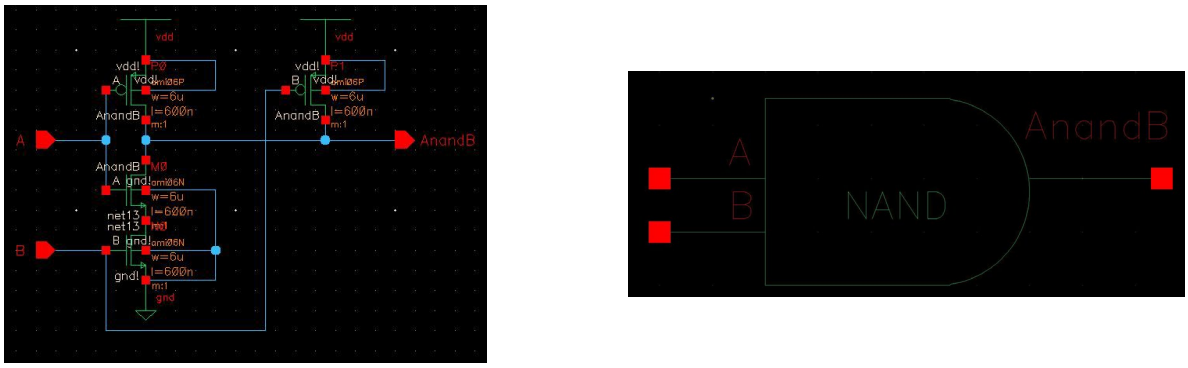


Figure 1. *Schematic (left) and symbol (right) of the CMOS NAND gate.*

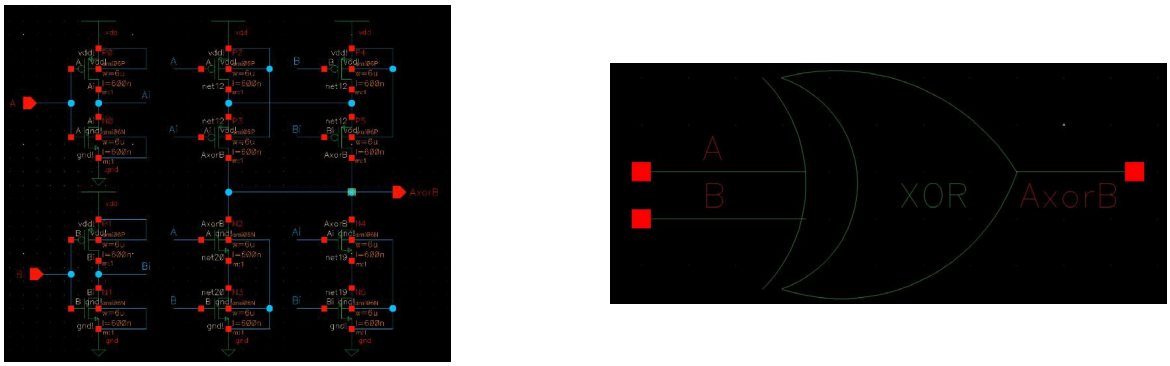


Figure 2. *Schematic (left) and symbol (right) of the CMOS XOR gate.*

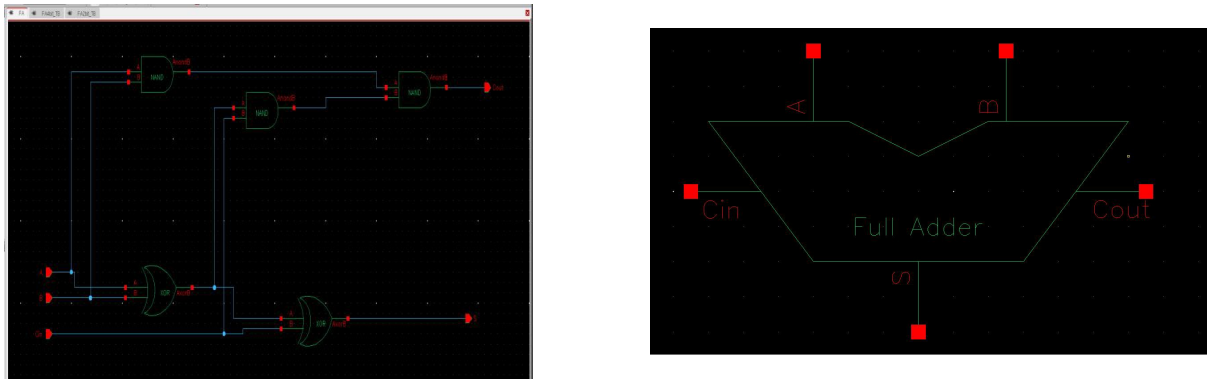


Figure 3. *Schematic (left) and symbol (right) of the CMOS full adder.*

2.1 Pre-layout functional simulation



Figure 4. *Pre-layout transient simulation of the CMOS full adder.*

Time (ns)	A	B	C_{in}	C_{out}	S	Status
0–30	1	1	1	1	1	Passed
31–50	1	1	0	1	0	Passed
51–75	1	0	1	1	0	Passed
76–100	1	0	0	0	1	Passed
101–125	0	1	1	1	0	Passed
126–150	0	1	0	0	1	Passed
151–180	0	0	1	0	1	Passed
185–200	0	0	0	0	0	Passed

Conclusion: All eight input combinations match the theoretical truth table — the pre-layout schematic is functionally correct.

2.2 Transient analysis — rise/fall time and propagation delay

Transient analysis was run on both outputs to extract rise time (t_r), fall time (t_f) and the two propagation delays (t_{pHL}, t_{pLH}). The average propagation delay is $\tau_p = (t_{pLH} + t_{pHL}) / 2$.

Carry-out

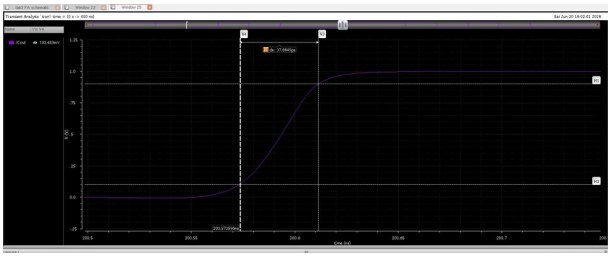


Figure 5. *C_{out} rise time t_r = 37.8845 ps.*

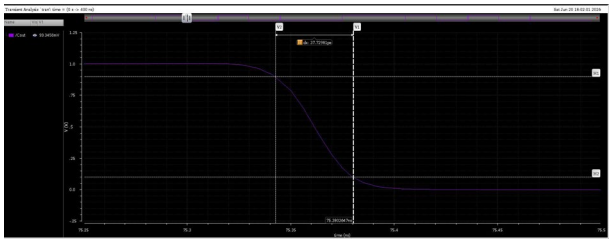


Figure 6. *C_{out} fall time t_f = 37.7298 ps.*

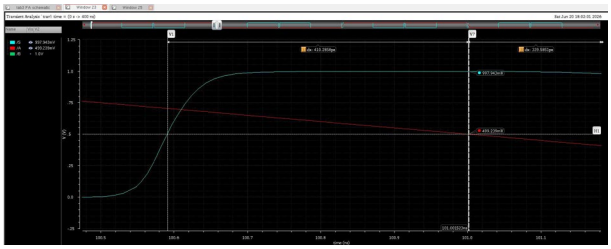


Figure 7. *C_{out} propagation delay t_{pHL} = 410.2 ps.*

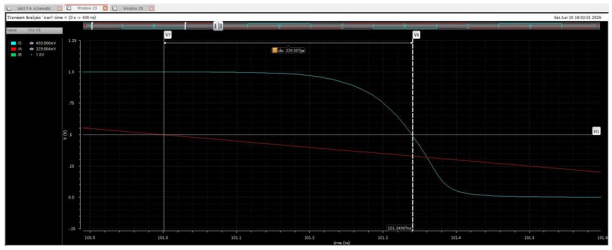


Figure 8. *C_{out} propagation delay t_{pLH} = 339.507 ps.*

Sum (S)

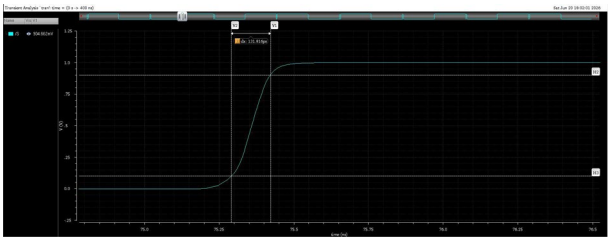


Figure 9. *S rise time t_r = 131.81 ps.*

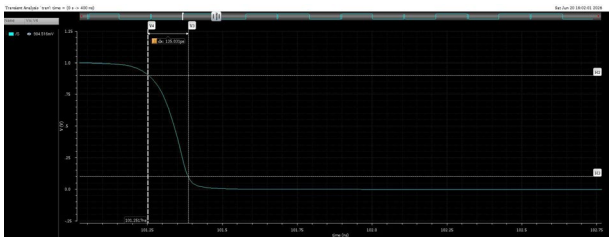


Figure 10. *S fall time t_f = 135.031 ps.*

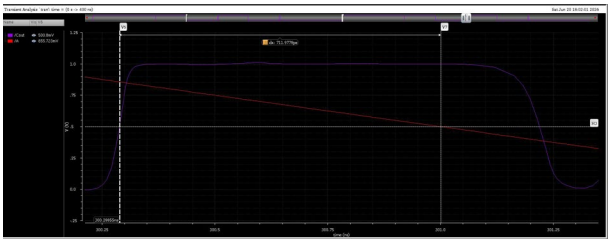


Figure 11. *S propagation delay t_{pHL} = 711.9776 ps.*

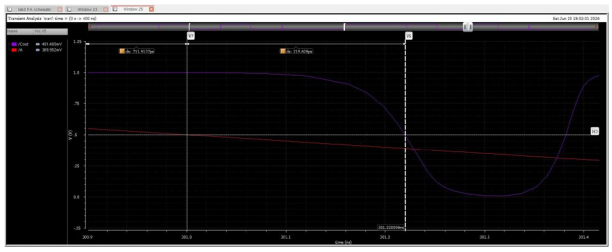


Figure 12. *S propagation delay t_{pLH} = 219.626 ps.*

Table 1 — Extracted timing summary of the 1-bit CMOS full adder

Output	t_r (ps)	t_f (ps)	t_{pHL} (ps)	t_{pLH} (ps)	$\tau_p = (t_{pLH}+t_{pHL})/2$
C_{out}	37.8845	37.7298	410.2	339.507	374.8535 ps
S	131.81	135.031	711.9776	219.626	465.8018 ps

3 PART B — LAYOUT DESIGN

Each cell was laid out with compact routing and shared diffusion to minimise area and parasitics, then verified with DRC and LVS.

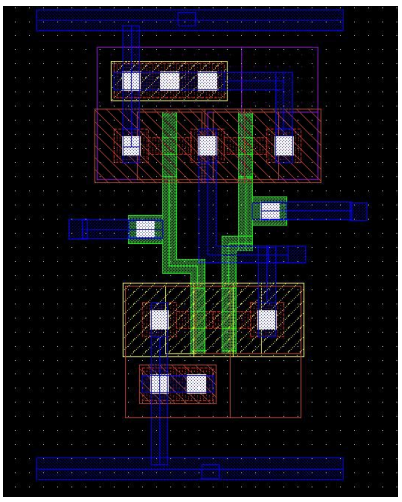


Figure 13. *NAND gate layout.*

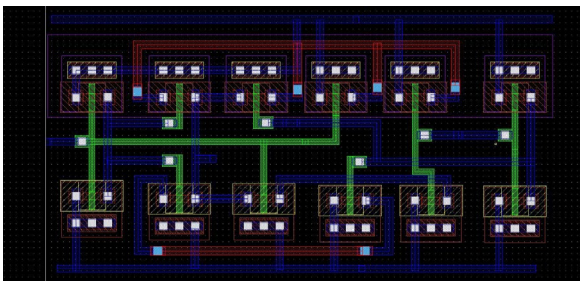
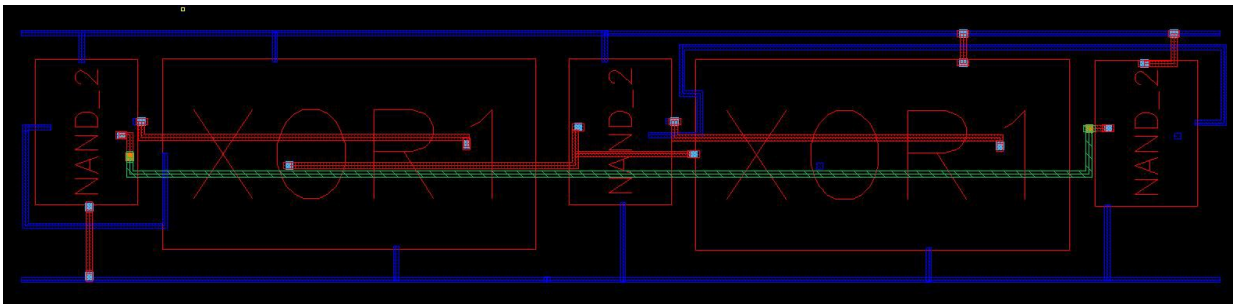


Figure 14. *XOR gate layout.*



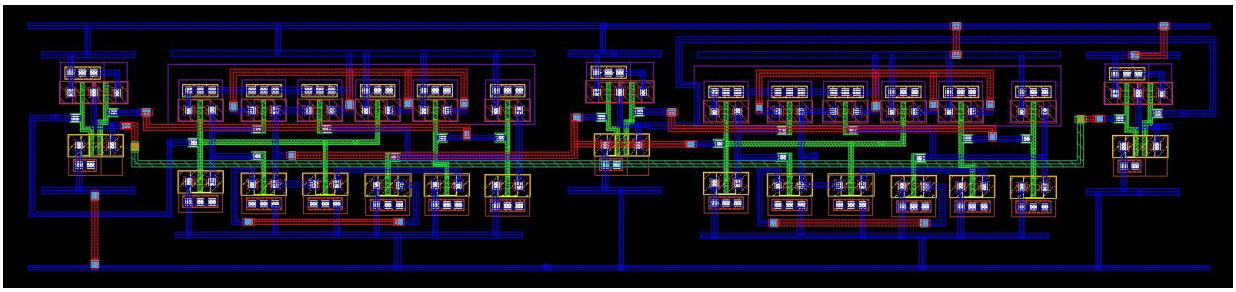


Figure 15. *Full-adder layout design.*

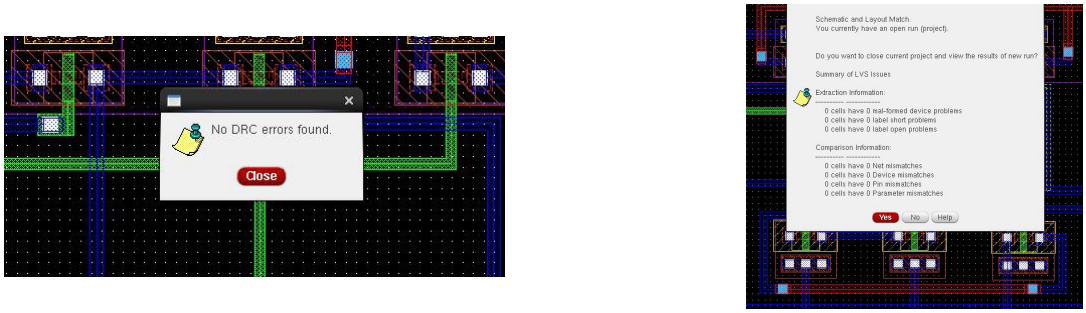


Figure 16. *DRC (left) and LVS (right) verification results.*

> **Conclusion:** The layout passes DRC with 0 errors and matches the schematic connectivity and device sizing under LVS. Compact routing and diffusion sharing keep the parasitic footprint and cell area to a minimum.

4 PART C — POST-LAYOUT SIMULATION



Figure 17. *Post-layout simulation of the CMOS full adder.*

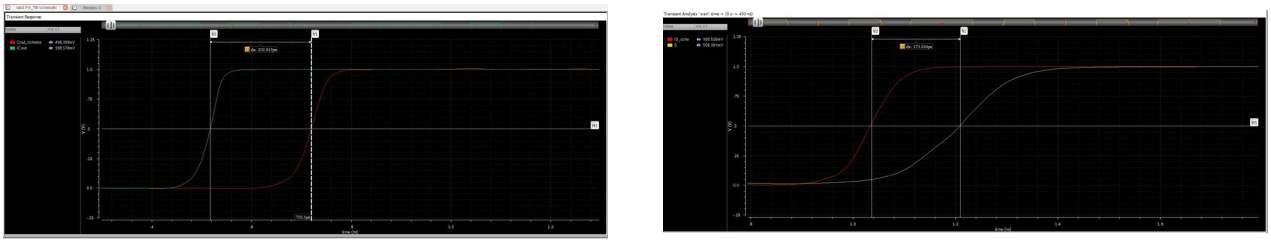


Figure 18. *Pre-layout vs. post-layout comparison of the full adder outputs.*

Output	Additional delay after layout extraction
C_{out}	≈ 202.82 ps
S	≈ 173.02 ps

Conclusion: Figure 18 shows a clear divergence between the pre- and post-layout waveforms: C_{out} slows by ≈202.82 ps and S by ≈173.02 ps. The extra delay comes from the parasitic RC of physical interconnect, routing layers and shared diffusion regions — confirming that careful layout optimisation is indispensable in high-frequency CMOS design.

5 EXERCISE 1 — INDIVIDUAL GATE VERIFICATION

Each primitive gate was designed and simulated separately before being used in the adder. For every gate the schematic, its transient response and the corresponding functional check are grouped together below.

5.1 CMOS inverter

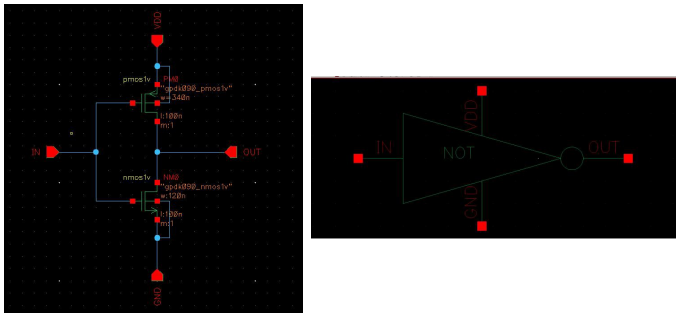


Figure 19. *Inverter schematic and symbol.*

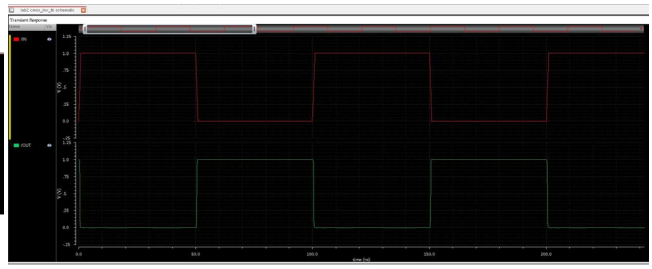


Figure 20. *Inverter transient simulation.*

Time (ns)	IN	OUT	Operation	Status
0–50	1	0	¬1 = 0	Passed
51–100	0	1	¬0 = 1	Passed

5.2 CMOS NAND gate

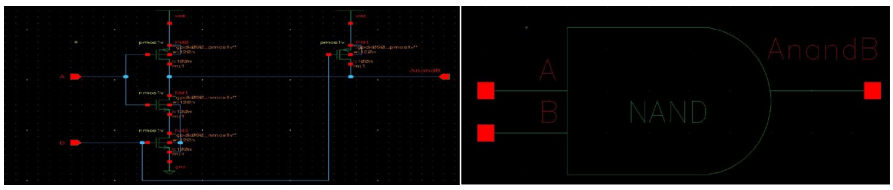


Figure 21. *CMOS NAND gate schematic and symbol.*

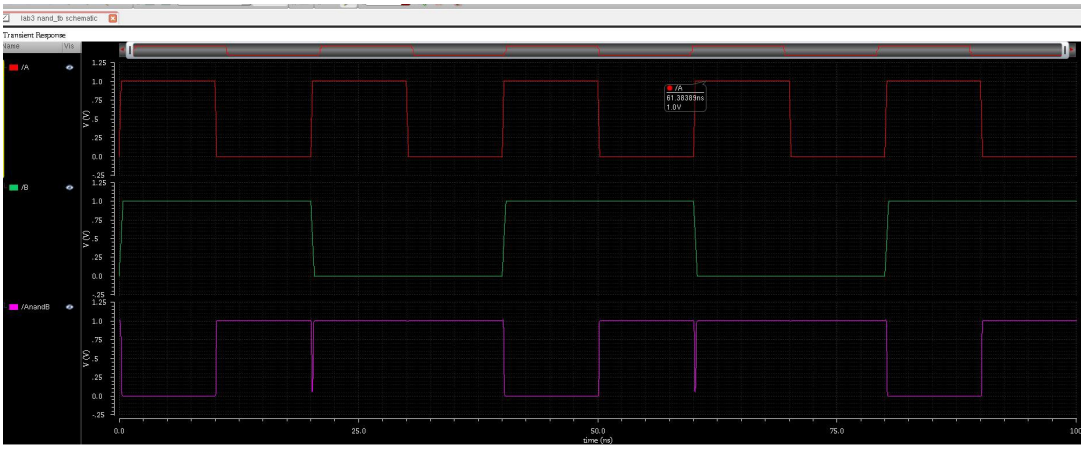


Figure 22. *CMOS NAND gate transient simulation.*

Time (ns)	A	B	OUT	Operation	Status
0–12.5	1	1	0	¬(1·1) = 0	Passed
≈13–20	0	1	1	¬(0·1) = 1	Passed
21–30	1	0	1	¬(1·0) = 1	Passed
31–40	0	0	1	¬(0·0) = 1	Passed

5.3 CMOS NOR gate

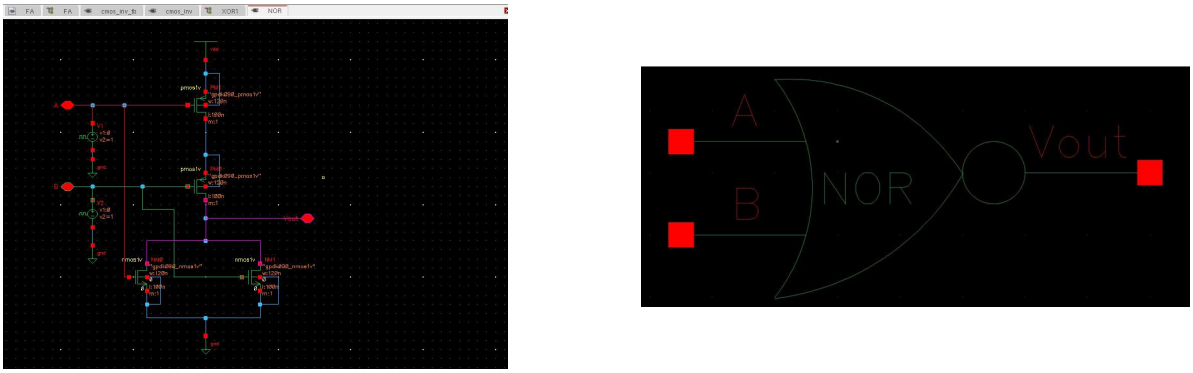


Figure 23. *CMOS NOR gate schematic (left) and symbol (right).*



Figure 24. *CMOS NOR gate transient simulation.*

Time (ns)	A	B	OUT	Operation	Status
0–5	1	1	0	¬(1+1) = 0	Passed
6–11	0	1	0	¬(0+1) = 0	Passed
12–17	1	0	0	¬(1+0) = 0	Passed
18–23	0	0	1	¬(0+0) = 1	Passed

5.4 CMOS XOR gate

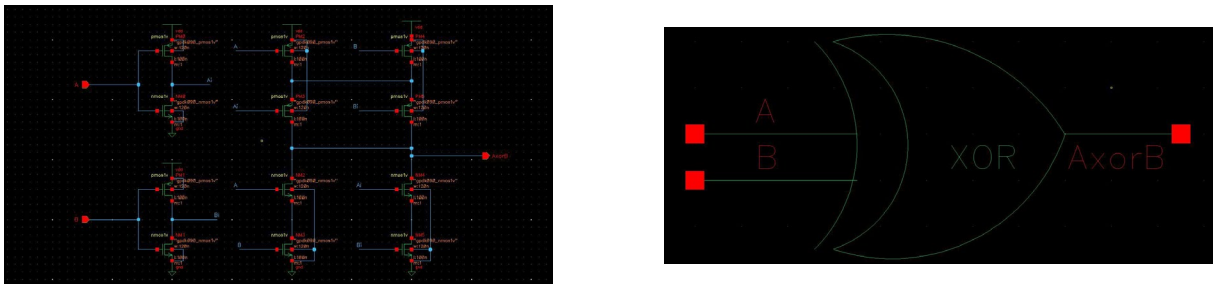


Figure 25. *CMOS XOR gate schematic (left) and symbol (right).*



Figure 26. *CMOS XOR gate transient simulation.*

Time (ns)	A	B	OUT	Operation	Status
≈0–25	1	1	0	$1 \oplus 1 = 0$	Passed
26–36	1	0	1	$1 \oplus 0 = 1$	Passed
37–50	0	0	0	$0 \oplus 0 = 0$	Passed
51–70	0	1	1	$0 \oplus 1 = 1$	Passed

Table 2 — Transistor count per logic gate

Logic gate	PMOS	NMOS	Total transistors
Inverter	1	1	2
NAND	2	2	4
NOR	2	2	4
XOR	6	6	12

6 EXERCISE 2 — MULTI-BIT ADDER EXTENSION

Several 1-bit full adders were chained into ripple-carry adders to study how area and delay scale with bit-width.

6.1 2-bit ripple carry adder

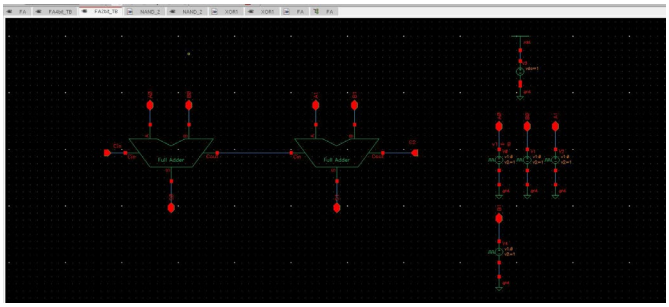


Figure 27. *2-bit ripple carry adder schematic.*

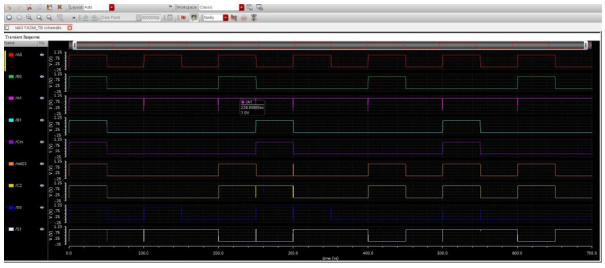


Figure 28. *Simulation of the 2-bit ripple carry adder.*

Area: A 2-bit RCA needs two full-adder blocks, so silicon area scales linearly with bit-width: 2 × 36 = **72 transistors**, i.e. double the 1-bit footprint.

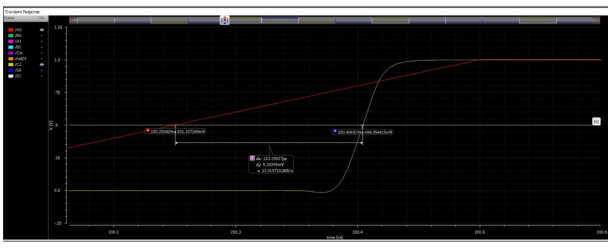


Figure 29. C_{out} *propagation delay* t_{pHL} *= 153.39 ps.*

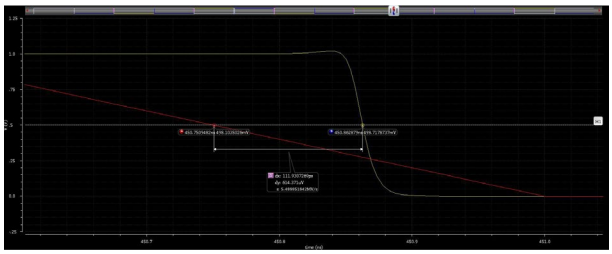


Figure 30. C_{out} *propagation delay* t_{pLH} *= 111.93 ps.*

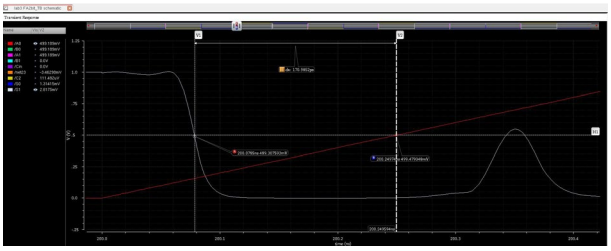


Figure 31. *S_1 propagation delay t_{pLH} = 170.98 ps.*

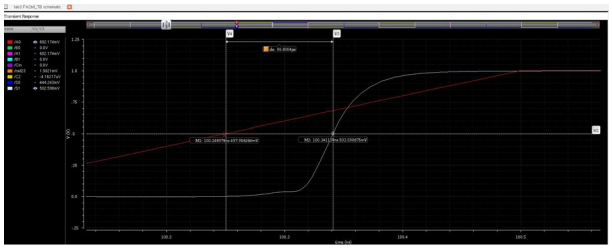


Figure 32. *S_1 propagation delay t_{pHL} = 90.8 ps.*

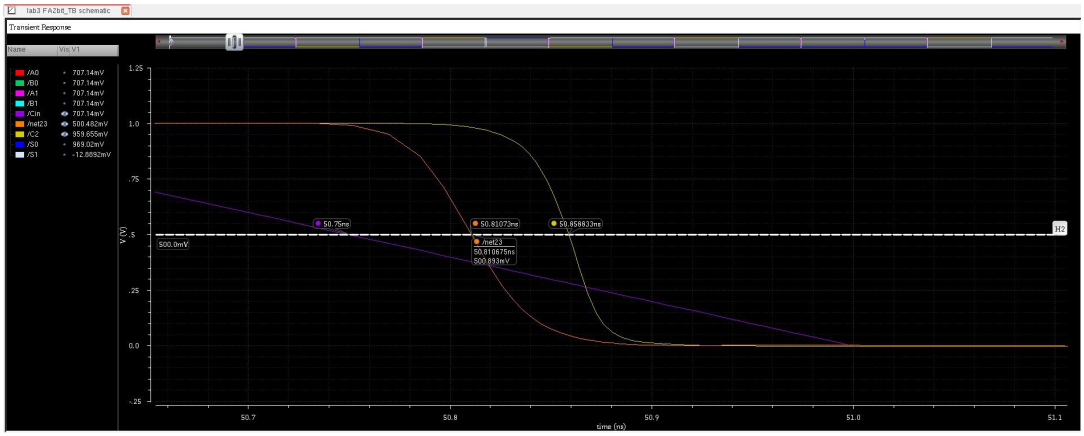


Figure 33. *Carry propagation effect in the 2-bit ripple carry adder.*

Table 3 — Extracted timing of the 2-bit ripple carry adder

Signal	t_{pHL} (ps)	t_{pLH} (ps)	τ_p (ps)
C_{out} (Fig. 29–30)	153.39	111.93	132.66
S_1 (Fig. 31–32)	90.8	170.98	130.89

Conclusion: Chaining two adder blocks preserves full functional integrity. The worst-case latency is 132.66 ps on C_{out} (Figures 29 & 30) and 130.89 ps on the most-significant sum bit S_1 (Figures 31 & 32). Figure 33 shows the carry transit time between consecutive stages varying between 40 ps and 60 ps, i.e. a mean stage penalty of 50 ps — confirming that the upper bits cannot settle until the carry has rippled through the preceding blocks.

6.2 4-bit ripple carry adder

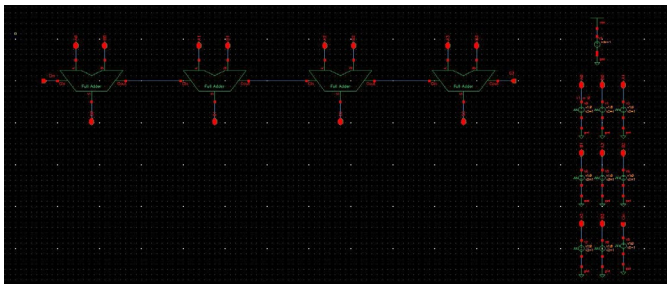


Figure 34. *4-bit ripple carry adder schematic.*



Figure 35. *Simulation of the 4-bit ripple carry adder.*

Area: Scaling stays predictable: the full 4-bit design integrates 144 MOSFETs, four times the base-cell silicon area.

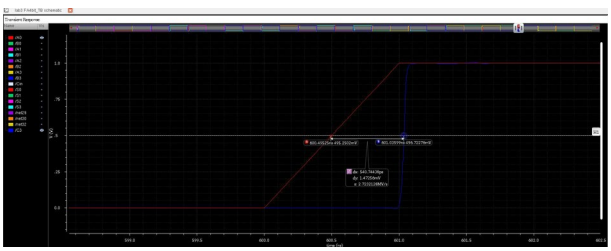


Figure 36. *C_3 propagation delay t_{pHL} = 540.74 ps.*

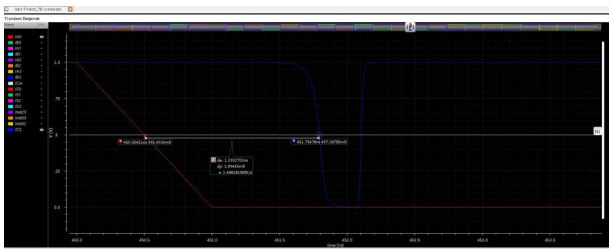


Figure 37. *C_3 propagation delay t_{pLH} = 1.29 ns.*

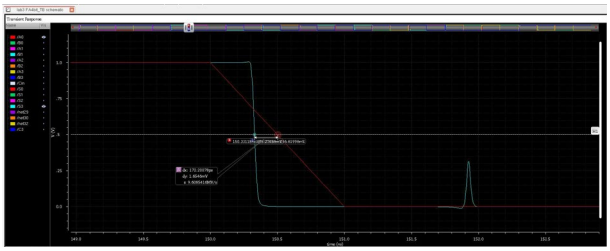


Figure 38. *S_3 propagation delay t_{pLH} = 172.2 ps.*

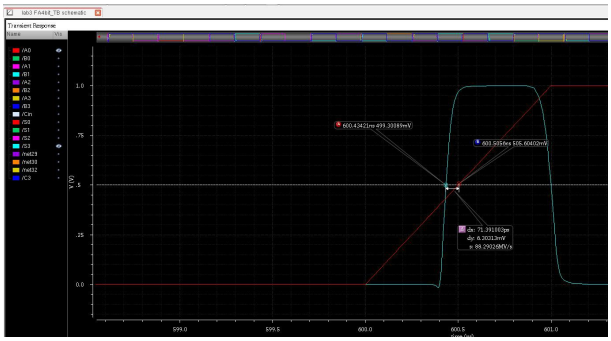


Figure 39. *S_3 propagation delay t_{pHL} = 71.39 ps.*

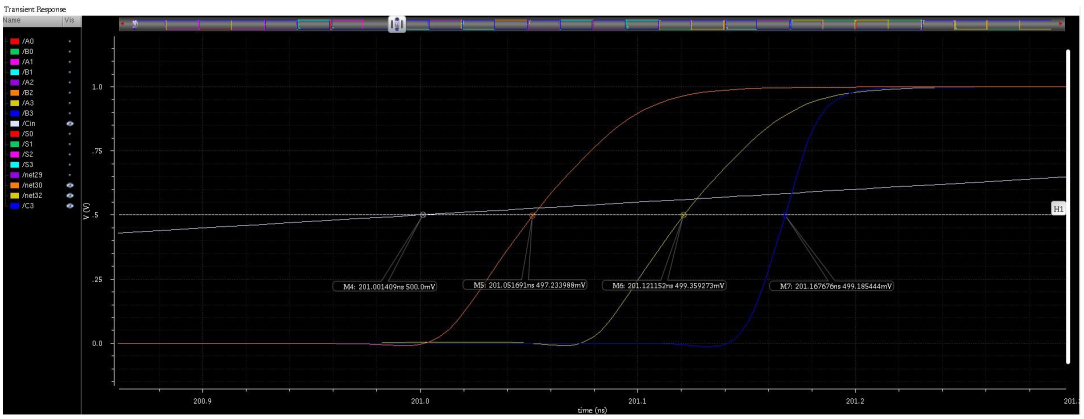


Figure 40. *Carry propagation effect in the 4-bit ripple carry adder.*

Table 4 — Extracted timing of the 4-bit ripple carry adder

Signal	t_{pHL} (ps)	t_{pLH} (ps)	τ_p (ps)
C_3 (Fig. 36–37)	540.74	1290	915.37
S_3 (Fig. 38–39)	71.39	172.2	121.795

Conclusion: The 4-bit configuration exposes the cost of cascading. The critical path stretches to 915.37 ps on C_{out} (Figures 36 & 37), while the most-significant sum bit S_3 needs 121.795 ps (Figures 38 & 39). Figure 40 shows the stage-to-stage carry transit ranging from 46.52 ps to 69.46 ps (mean ≈ 55.42 ps), and the silicon area grows in direct proportion to the number of full-adder cells. Despite its structural simplicity and area efficiency, the serial carry chain of the RCA is therefore a severe bottleneck on the maximum achievable clock frequency.

6.3 Overall scaling summary

Table 5 — Area and delay scaling of the ripple-carry architecture

Design	Full-adder cells	Transistors	Worst-case carry delay	Mean carry delay per stage
1-bit full adder	1	36	374.8535 ps (C_{out})	—
2-bit RCA	2	72	132.66 ps (C_{out})	≈ 50 ps
4-bit RCA	4	144	915.37 ps (C_3)	≈ 55.42 ps

Area grows linearly with bit-width (36 transistors per cell), while the carry path grows serially — the fundamental area/speed trade-off of the ripple-carry architecture.